

Filter-matched rate calibration for variable-rate PING training

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Abstract

This staged experiment tests which Poisson rates preserve MNIST class evidence. An artificial neural network (ANN) is planned to classify static features formed by passing each pixel's spike train through an AMPA synapse and non-spiking membrane, then averaging its voltage over the presentation.

ANN accuracy against rate remains the planned primary result, but no decoder was trained here. An explicitly authorized exploratory continuation instead tested a linear transfer-function prediction and constructed complete sampled MNIST feature images from the preserved empirical library.

Step 1 is complete. The local generator passed all 8 focused checks, including agreement with the shared `tools/snn` cell below the registered numerical tolerance. The original Step 2 pilot stopped at $K = 512$; an explicitly authorized extension then stabilized both empirical moments at $K = 2048$ under the unchanged tolerances. The complete empirical library was then generated, but 5 of 9180 adjacent-intensity comparisons exceeded the locked Monte Carlo tolerance. Step 2 therefore failed its original gate. That failure remains unchanged. Under the later amendment, Step 3 found that the stationary linear model overpredicted finite-window variance, especially at low drive and larger conductance. Step 4 reproduced pooled pixel statistics and recognizable feature structure, but failed its locked low-rate image-level checks. Steps 5–7 remain pending.

Purpose and scope

The question is:

At a given encoding rate, does the temporally filtered image still contain enough information for an ANN to recognize the digit?

The target PING architecture has 784 pixel channels feeding conductance-based excitatory cells. Its fixed synaptic and membrane equations define the features, without trained input, recurrent, or output weights. The ANN thus measures filtered-input decodability independently of a trained PING network.

The experimental logic is:

1. Convert 0.25–25 Hz Poisson inputs into static, filter-matched features.
2. Measure their distributions and test a linear variance prediction.
3. Train the ANN across the rate grid with the held-out test set sealed.

4. Measure ANN accuracy and locate its lowest decodable rates.

- Nonlinear ANN above chance: this decoder can access class evidence.
- Nonlinear ANN at chance: this representation and decoder cannot extract reliable evidence.
- Nonlinear ANN succeeds but linear decoder fails: the evidence is not linearly accessible.

The ANN threshold is a decoder-relative decodability edge, not an absolute information boundary or a prediction of PING accuracy. The practical training floor instead applies a predeclared useful-accuracy criterion. A later experiment must test whether PING can train across the selected range.

Methods

1. **Generated and validated filter-matched pixel features.** Every image used a 200 ms presentation and 0.1 ms timestep. The tested rates are 0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 4, 5, 10, 25 Hz; dense sampling below 5 Hz resolves the lower edge, while 25 Hz is the proposed later training ceiling. Training samples uniformly from these points. Seeds 42, 43, 44 define independent feature, response-library, and ANN runs.

`protocol.json` records the grid and deterministic MNIST train, validation, and held-out indices. Pixel-intensities x remain in $[0, 1]$. If r is the encoding rate at pixel-intensity one, the expected pixel rate is $r x$.

Each pixel drives an independent AMPA synapse and uncoupled, non-spiking excitatory membrane, capturing AMPA decay, conductance-dependent integration, and within-window timing that raw counts omit. This probe has no threshold, reset, recurrence, or trained weights. It uses the target PING cell’s decay-then-add synapse and exponential-Euler membrane update, implemented locally without changing the shared simulator.

The complete feature path is

$$S_i(t) \rightarrow g_i^{\text{pix}}(t) \rightarrow v_i(t) \rightarrow z_i \rightarrow \text{ANN}. \quad (1)$$

Here S_i , g_i^{pix} , v_i , and z_i are pixel i ’s spike train, AMPA conductance, subthreshold voltage, and time-averaged feature; t is time and ANN is the classifier.

At each timestep, we drew the encoded input according to

$$S_i(t) \sim \text{Bernoulli}(r \Delta t x_i). \quad (2)$$

Here x_i is grayscale pixel-intensity, r is the encoding rate in spikes per second at pixel-intensity one, Δt is the timestep in seconds, and Bernoulli is an independent binary draw. Configurations with probability above one are invalid.

We updated the AMPA conductance using

$$g_i^{\text{pix}}(t) = \beta_{\text{AMPA}} g_i^{\text{pix}}(t-1) + w_{\text{probe}} S_i(t). \quad (3)$$

$$\beta_{\text{AMPA}} = \exp\left(-\frac{\Delta t}{\tau_{\text{AMPA}}}\right). \quad (4)$$

Here β_{AMPA} is one-timestep conductance retention, τ_{AMPA} is AMPA decay time, w_{probe} is conductance added per spike, and $\exp$ is the exponential function. Other symbols follow

Equation 2. The primary 1.2 μS probe is the target architecture’s nominal mean initial pixel-to-excitatory conductance; 0.6 and 2.4 μS test half and double scale.

We integrated the membrane without thresholding or resetting:

$$C_E \frac{dv_i}{dt} = g_{L,E}(E_L - v_i) + g_i^{\text{pix}}(t)(E_e - v_i). \quad (5)$$

$$\tau_{\text{eff},i}(t) = \frac{C_E}{g_{L,E} + g_i^{\text{pix}}(t)}. \quad (6)$$

Here C_E is excitatory capacitance; $g_{L,E}$ is leak conductance; E_L and E_e are leak and AMPA reversal potentials; and $\tau_{\text{eff},i}$ is the instantaneous conductance-dependent time constant. Other symbols follow Equation 1.

We initialized voltage at E_L and conductance at zero, simulated one complete presentation, and calculated

$$z_i = \frac{1}{T} \int_0^T (v_i(t) - E_L) dt. \quad (7)$$

Here T is presentation duration and z_i is mean baseline-subtracted voltage. We did not divide by rate or disclose it to either decoder, so lower rates retain their weaker, noisier signal.

Focused tests covered encoder probability, count moments, pixel independence, AMPA decay, exponential-Euler voltage updates, deterministic replay, agreement with one uncoupled target-PING excitatory cell, and spike-timing sensitivity. All 8 checks passed. At the seeded count validation point, the expected and observed mean counts were 4 and 4.01 the largest absolute inter-channel count correlation was 0.023.

The corresponding plot is defined in Results, Step 1.

2. **Generated and validated the empirical response library.** Before generating a final library, we locked candidate draw counts K of 64, 128, 256, and 512. None passed. After recording that failure, the user explicitly authorized an extension at $K = 1,024$ and 2,048 without changing the convergence rule. The deterministic pilot covered 270 conditions: six intensities, five rates, all three probe conductances, and all three registered seeds. For each candidate, we compared the first K draws with the next K independent draws.

Mean and unbiased-variance discrepancies were divided by locked absolute or relative tolerances, whichever was larger. Both metrics had to keep their 95th percentile at or below 1 and their maximum at or below 2. The smallest passing K would have selected the final draw count. If none passed at the maximum, the protocol required stopping without changing the rule.

The extension selected $K = 2048$. We then ran that many independent Step 1 draws for every grayscale level, registered rate, probe conductance, and seed. The retained float32 array has shape $3 \times 3 \times 12 \times 256 \times 2048$ in the ordered axes seed, probe conductance, rate, intensity, and draw. Its payload occupies 226492416 bytes in local scratch and is authenticated by the SHA-256 digest `#m.library_sha256`.

The planned final library would estimate the conditional mean

$$\hat{\mu}_z(x, r, w_{\text{probe}}) = \frac{1}{K} \sum_{k=1}^K z^k. \quad (8)$$

and conditional variance

$$\hat{\sigma}_z^2(x, r, w_{\text{probe}}) = \frac{1}{K-1} \sum_{k=1}^K (z^k - \hat{\mu}_z)^2. \quad (9)$$

Here x , r , and w_{probe} are pixel-intensity, rate, and probe conductance; $z^{(k)}$ is draw k ; K is the draw count; and Equations 8 and 9 estimate the conditional mean and variance.

We retained all K simulated values as the primary empirical response library. During a later ANN feature-generation stage, the registered design would draw one value according to

$$J \sim \text{DiscreteUniform}(1, \dots, K), \quad z_{\text{sample}} = z^J. \quad (10)$$

Here J is uniform on the K draws and z_{sample} is the sampled feature; other symbols follow Equations 8 and 9. Mean and variance summarize noise, but the ANN samples the retained empirical values because low-rate responses may be zero-heavy, skewed, and non-Gaussian.

The generator wrote 96 deterministic, independently authenticated intensity chunks. Exact replay passed for one predeclared chunk from every seed. The complete grid, finite values, physical bounds, zero-intensity resting response, independent streams, independently recomputed moments, fresh direct simulations, and float32 fidelity all passed their checks. The low-rate representative condition had zero fraction 0.986, skewness 8.666, and 85 distinct retained values, preserving its discrete, non-Gaussian structure.

The required monotonicity check did not pass. It compared every adjacent intensity and rate pair using the predeclared 3-standard-error tolerance. 5 of 9180 intensity pairs exceeded it, while 0 of 8448 rate pairs did. The rule was not weakened after inspection, so this completed library is a preserved killed attempt rather than a validated input to Step 3.

The corresponding plot is defined in Results, Step 2.

3. **Compare the empirical variance with a linear-filter prediction.** This supplementary check asks whether a local linear approximation explains the Step 2 feature variance. It will not set the training range.

For mean spike rate λ , stationary mean conductance is

$$|g|_\lambda = \lambda w_{\text{probe}} \tau_{\text{AMPA}}. \quad (11)$$

and stationary mean voltage is

$$|v|_\lambda = \frac{g_{\text{L,E}} E_L + |g|_\lambda E_e}{g_{\text{L,E}} + |g|_\lambda}. \quad (12)$$

Here λ is encoding rate times pixel-intensity; $\text{bar}(g)_\lambda$ and $\text{bar}(v)_\lambda$ are stationary mean conductance and voltage. Other symbols follow Step 1.

Linearizing the synapse and membrane around that operating point gives

$$G_{\lambda(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - |v|_{\lambda}}{i\omega C_E + g_{\text{L,E}} + |g|_{\lambda}}. \quad (13)$$

Here $G_{\lambda}(\omega)$ is the local synapse-plus-membrane transfer function; ω is angular frequency; and i is the imaginary unit. Because mean conductance changes gain and effective membrane time constant, Equation 13 defines a family of responses.

The finite averaging window contributes

$$A_{T(\omega)} = \frac{1 - \exp(-i\omega T)}{i\omega T}. \quad (14)$$

$$H_{\lambda(\omega)} = A_{T(\omega)} G_{\lambda(\omega)}. \quad (15)$$

Here $A_T(\omega)$ is the duration- T averaging response; $H_{\lambda}(\omega)$ is the complete response from input fluctuation to averaged feature; T is presentation duration; and $\exp$ is the exponential function.

For centred ideal Poisson input,

$$S_{\text{in}(\omega)} = \lambda. \quad (16)$$

$$S_{z(\omega)} = |H_{\lambda(\omega)}|^2 S_{\text{in}(\omega)}. \quad (17)$$

$$\text{Var}_{\text{linear}(z)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |H_{\lambda(\omega)}|^2 S_{\text{in}(\omega)} d\omega. \quad (18)$$

Here $S_{\text{in}}(\omega)$ and $S_z(\omega)$ are input and predicted output power spectral densities; $|H_{\lambda}(\omega)|^2$ is transmitted noise power; $\text{Var}_{\text{linear}}(z)$ is predicted feature variance; and π is the circle constant. Appendix A derives Equations 11–18.

Under a timestamped exploratory amendment, we evaluated every distinct calibration point, compared predicted with empirical Step 2 variance, and validated low-, middle-, and high-drive gains by sinusoidally modulating the numerical probe. The amendment did not relabel the original Step 2 gate. This stationary, continuous-time, linearized Poisson model approximates the finite, discrete Bernoulli simulation and remains diagnostic only.

The corresponding plot is defined in Results, Step 3.

4. **Constructed and validated complete feature images.** We used only the official 60,000-image MNIST training partition. Indices 0–54,999 and 55,000–59,999 remain the locked future decoder-training and validation partitions; the official test partition was not loaded. For each uint8 pixel, the sampler used its exact 0–255 intensity index and selected one of the $K = 2048$ authenticated empirical draws with independent, deterministic pixel and image streams.

The direct comparison used training images 0–15, 28×28 pixels, eight independent replicates, all three conductances, and rates 0.25, 3, and 25 Hz. We compared pooled pixel moments, image-level moments, zero fractions, absolute and relative differences, and the spatial correlation of per-pixel means. The thresholds and streams were locked before

outcomes. This validation was required before ANN training; its low-rate checks did not all pass, so no decoder followed.

5. **Train the mixed-rate decoders.** We will train a primary ANN with 784 inputs, one 1,024-unit rectified-linear hidden layer, and ten outputs. The ANN learns both weight matrices from voltage features. A regularized linear softmax decoder on the same images tests linear accessibility. Neither model receives the rate or pretrained weights.

Seeds 42, 43, 44 use Adam, learning rate 0.001, batch size 256, and initially at most 15 epochs. Every presentation samples a rate uniformly and regenerates features. Validation alone controls model selection, regularization, early stopping, and any epoch extension; test data remain sealed until Step 6. We will record each configuration, selected epoch, and training history.

The primary ensemble uses the 1.2 μS probe; separate 0.6 and 2.4 μS runs test sensitivity. Conductance conditions are neither pooled nor disclosed.

The corresponding plot is defined in Results, Step 5.

6. **Generate inference-only psychometric curves and choose two rate thresholds.** We will freeze all decoders before testing. At every tested rate, seeds use the same held-out images and fixed feature draws; additional draws measure encoding variability. The primary curve is

$$A_r(r) = P(\text{correct} \mid r, \text{mixed-rate nonlinear decoder}). \quad (19)$$

Here A_r is held-out nonlinear-ANN accuracy, P is correct-classification probability, and r is rate. The linear decoder is diagnostic only.

Bootstrap resampling of held-out images, probe draws, response-library simulations, and decoder seeds gives lower confidence bound L_r . The decoder-relative edge is

$$r_{\text{decode}} = \text{lowest } r \in \mathcal{R} \text{ satisfying } L_r(r) > \frac{1}{N_{\text{class}}}. \quad (20)$$

The practical training floor is

$$r_{\text{train}} = \text{lowest } r \in \mathcal{R} \text{ satisfying } L_r(r) \geq a_{\text{use}}. \quad (21)$$

Here $\mathcal{R}$ is the tested grid; N_{class} is the number of classes; r_{decode} is the lowest rate reliably above chance; a_{use} is the predeclared 50% useful-accuracy target; and r_{train} is the lowest rate whose lower bound reaches it. The target is locked before testing.

Primary thresholds use the 1.2 μS nonlinear ANN. The linear decoder and 0.6 and 2.4 μS runs show sensitivity to decoder capacity and conductance. We will not treat interpolated rates as observations.

The corresponding plot is defined in Results, Step 6.

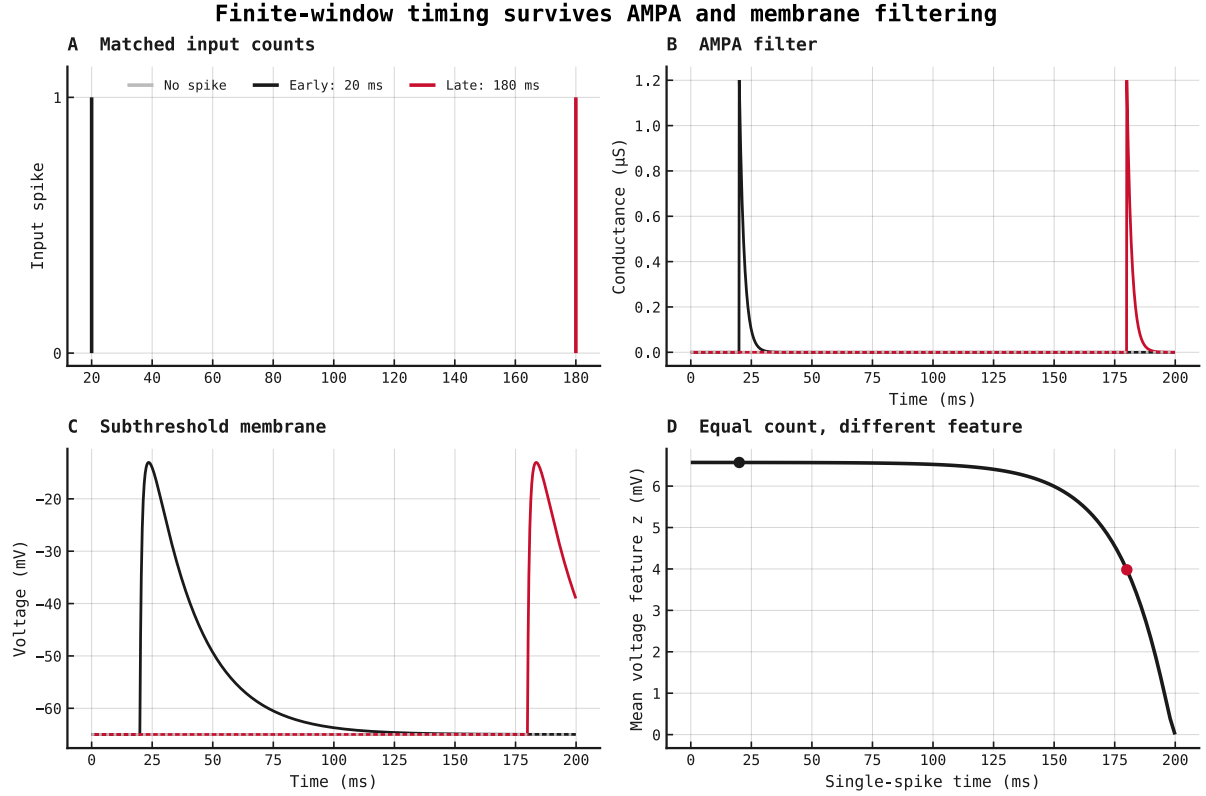
7. **Report the training-range decision and stop.** We will write `decision.json` with r_{decode} , r_{train} , their uncertainty, the probe-conductance sensitivity, decoder and artifact hashes, and all rule outcomes. The recommended later PING range is $r_{\text{train}} - 25$ Hz. If the floor shifts by more than one adjacent grid point across conductances, we will report

plausible floors rather than one value. This experiment stops at the recommendation; a separate experiment must train and test PING across it.

Results

Step 1 is complete. The Step 2 convergence pilot selected K after an authorized extension. The full library was generated, then Step 2 was killed by its required monotonicity validation. A timestamped post-hoc amendment subsequently authorized exploratory Steps 3–4 without changing that failure. Step 3 completed; Step 4 stopped at its locked low-rate image-level validation. Steps 5–7 remain pending.

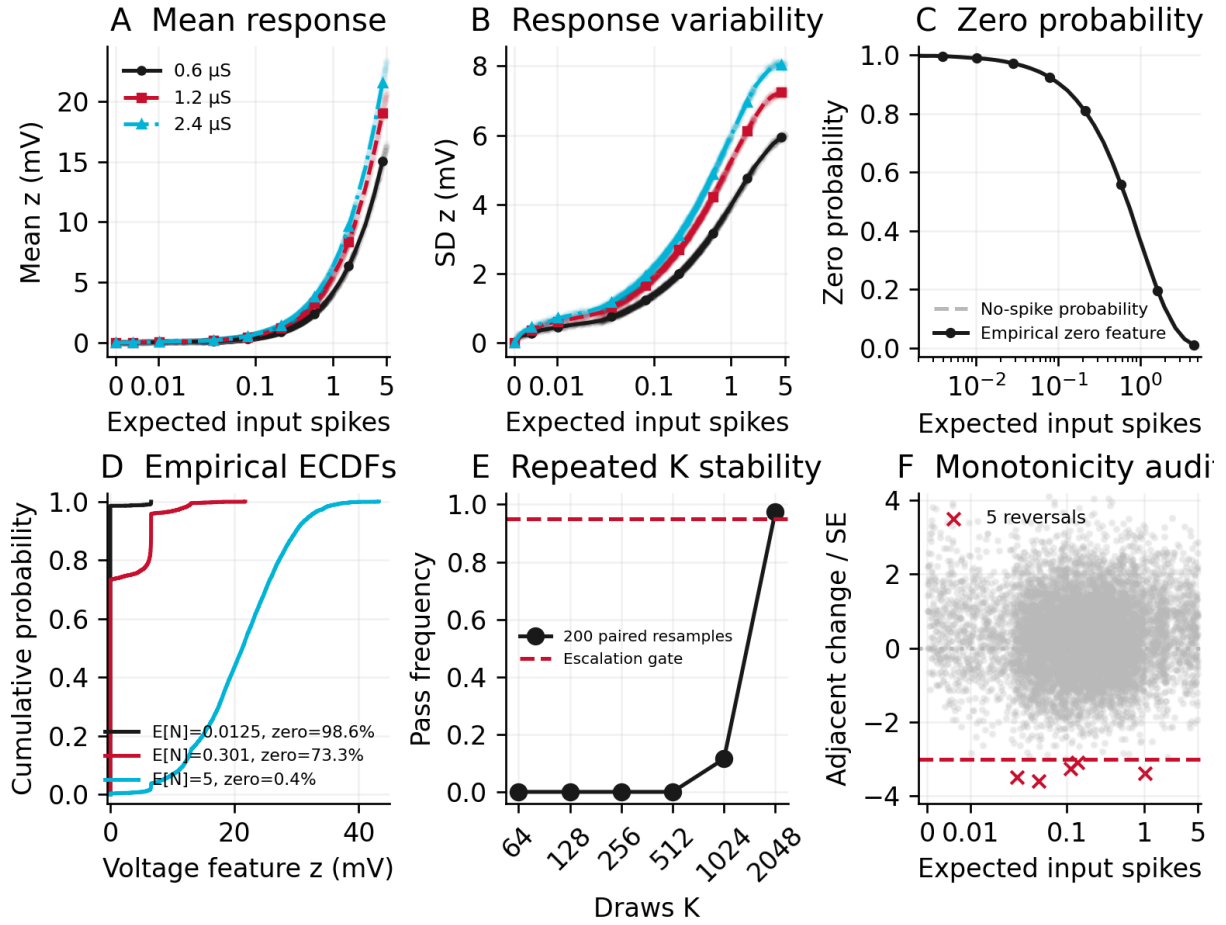
Step 1: filter-matched feature generation



Finite-window probe dynamics. Panels A–C show input spikes, AMPA conductance, and subthreshold voltage for no-spike, early-spike, and late-spike inputs. Panel D places the single spike across the presentation and reports the resulting mean voltage feature. With the same one-spike count, a spike at 20 ms produced $z = 6.571$ mV, whereas one at 180 ms produced 3.98 mV. Earlier input therefore contributed 2.591 mV more to the finite-window average.

The decay-then-add check reproduced the first two conductances with a maximum absolute error of 0 μS . The independently calculated exponential-Euler step and the complete local trajectory both matched the existing `tools/snn` uncoupled cell within their registered machine-precision tolerances. This validates the Step 1 feature generator only; it does not establish MNIST decodability or select a PING input-rate range.

Step 2: empirical response library

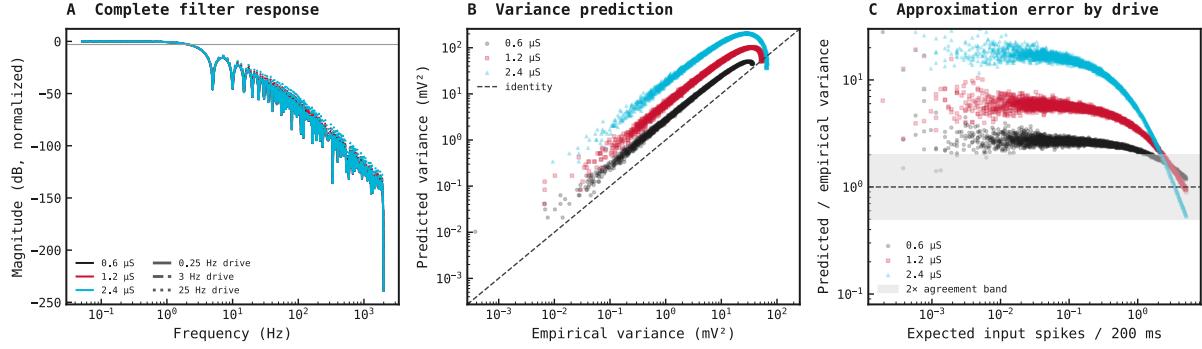


Complete empirical response library and validation evidence. Expected input spikes are encoding rate times normalized grayscale intensity times the 200 ms presentation. Panels A and B show conditional mean feature z and its standard deviation against that common drive variable. Faint points are all rate–intensity conditions; lines are binned medians, with colour, marker, and line style identifying probe conductance. Panel C compares the empirical probability of an exactly zero feature with the analytical no-spike probability. Panel D shows empirical cumulative distributions at low, transitional, and high drive for the nominal probe. Panel E reports the fraction of 200 paired bootstrap repetitions passing all four unchanged stability limits at each K ; its dashed line is the locked 0.95 decision threshold. Panel F expresses every adjacent-intensity mean change in units of its Monte Carlo standard error; its dashed line is the locked rejection boundary. The response changes smoothly with drive and retains its low-rate zero mass, but 5 isolated comparisons crossed that boundary, so Step 2 failed and no later stage ran.

The array contains 56623104 float32 values with shape $3 \times 3 \times 12 \times 256 \times 2048$ and a 226492416-byte payload. Zero intensity returned the resting feature exactly, direct simulation agreed at every predeclared condition, and exact chunk replay passed for all registered seeds. These checks establish an authenticated empirical library and its failure mode only. They do not establish MNIST decodability, a training-rate floor, or PING accuracy.

The post-hoc repeated-resampling diagnostic found that $K = 1024$ passed 23 of 200 repetitions, whereas $K = 2048$ passed 195, a frequency of 0.975. This exceeded the locked escalation threshold, so no independent $K = 4,096$ comparison was run. The diagnostic supports describing $K = 2048$ as typically stable under the registered tolerances, not as an asymptotic plateau.

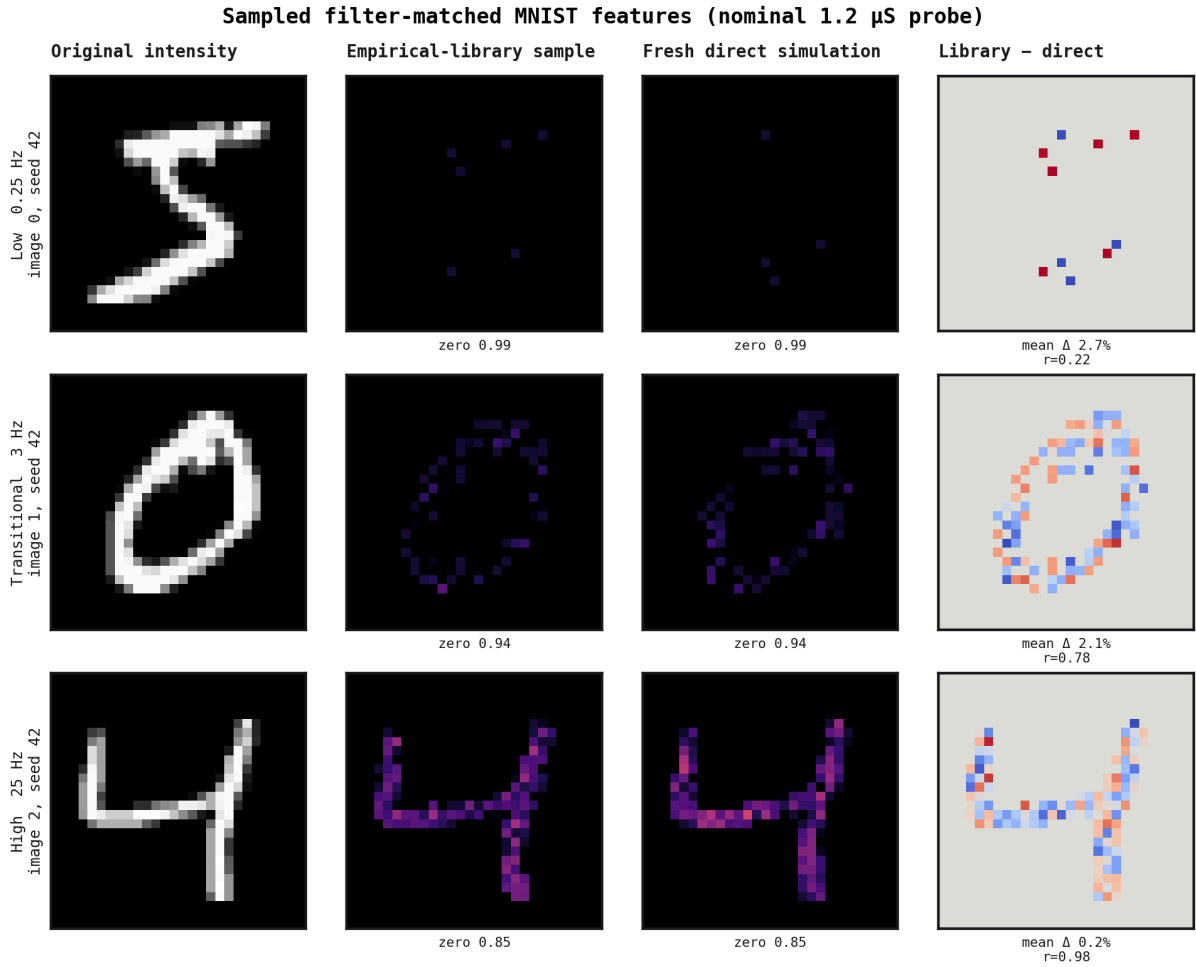
Step 3: linear-filter prediction



Stationary linear prediction versus the finite empirical probe. Panel A shows the synapse, membrane, and 200 ms averaging response at low, middle, and high drive; colour identifies probe conductance and line style identifies drive. Panel B compares all nonzero calibration points with identity. Panel C shows the predicted-to-empirical variance ratio against expected input spikes; the shaded band marks agreement within a factor of two. The approximation improves with drive but generally overpredicts the empirical finite-window variance.

All 27 numerical sinusoidal gain checks passed; the largest relative error was 0.0003. The maximum frequency-grid refinement change was 0.00000025, and the widened-bound change was 0.00000004, both below the locked 0.2% tolerance. At the nominal 1.2 μS probe, median predicted to empirical variance ratios were 5.78 at low drive, 5.014 at transitional drive, and 2.307 at high drive. The linear model is therefore a useful failure diagnostic, not a substitute for the empirical library.

Step 4: complete feature images



Empirical-library samples versus fresh direct Step 1 simulations. Rows show 0.25, 3, and 25 Hz at the nominal 1.2 μ S probe. Comparable voltage images use identical 0–65 mV limits. The low-rate row retains the registered zero-heavy, discrete structure; transitional and high rates preserve recognizable digit structure. The displayed samples are illustrative; the validation used all three conductances, 16 fixed images, and eight replicates per condition.

Six of nine probe–rate conditions passed every locked comparison. At the nominal probe, pooled-mean relative differences were 0.027, 0.021, and 0.002 from low to high drive; spatial correlations were 0.217, 0.78, and 0.977. All pooled pixel-moment and zero-fraction checks passed. The three low-rate conditions failed image-level mean and variance tolerances; the 2.4 μ S low-rate condition also reached correlation 0.198 against the locked 0.20 minimum. The thresholds and replicate count were not changed after inspection. Step 4 is therefore a preserved validation failure, and it establishes neither decodability nor classification accuracy.

Step 5: mixed-rate decoder training

Pending plot. Pending Step 5. Planned file: `decoder_training.svg`.

Planned caption. Mixed-rate training. Training and validation loss and accuracy are shown by epoch and seed for both decoders; a rate histogram checks uniform sampling. Held-out accuracy is excluded.

Step 6: ANN psychometric curve and thresholds

Pending plot. Pending Step 6. Planned file: `ann_psychometric.svg`.

Planned caption. Decoder-relative psychometrics at 200 ms, spanning 0.25–25 Hz with a 0.25–5 Hz inset. Panel A compares frozen nonlinear and linear decoders at 1.2 μS , with chance and both thresholds; B compares nonlinear accuracy at 0.6, 1.2, and 2.4 μS .

Uncertainty covers encoding and library draws and ANN seeds.

Step 7: handoff

Pending output. `decision.json` will report the decoder-relative edge, practical training floor, uncertainty, probe-conductance sensitivity, and the recommended rate range for a separate variable-rate PING training experiment.

Relation to prior work

Filtered conductance shot noise can violate fixed-time-constant models [1], and Gaussian approximations can miss skewed responses [2], supporting the empirical response library. Decoder performance is not absolute neural information [3], and linear and nonlinear decoders can differ [4], motivating a decoder-relative edge and both decoder diagnostics.

References

1. Wolff & Lindner: *Mean, Variance, and Autocorrelation of Subthreshold Potential Fluctuations Driven by Filtered Conductance Shot Noise*. Neural Computation, 2010. doi:10.1162/neco.2009.02-09-958
2. Brigham & Destexhe: *Nonstationary Filtered Shot-Noise Processes and Applications to Neuronal Membranes*. Physical Review E, 2015. doi:10.1103/PhysRevE.91.062102
3. Quian Quiroga & Panzeri: *Extracting Information from Neuronal Populations: Information Theory and Decoding Approaches*. Nature Reviews Neuroscience, 2009. doi:10.1038/nrn2578
4. Warland, Reinagel & Meister: *Decoding Visual Information From a Population of Retinal Ganglion Cells*. Journal of Neurophysiology, 1997. doi:10.1152/jn.1997.78.5.2336

Appendix A: linear-filter derivation

This appendix derives Equations 11–18 from the same AMPA synapse and subthreshold membrane used by the empirical probe. It replaces the finite Bernoulli drive with a continuous-time Poisson point process and linearizes the membrane around the mean drive at each calibration condition.

A.1 Stationary operating point

Represent pixel i 's idealized spike train and synaptic conductance by

$$s_{i(t)} = \sum_k \delta(t - t_k), \quad \frac{dg_{i(t)}}{dt} = -\frac{g_{i(t)}}{\tau_{\text{AMPA}}} + w_{\text{probe}} s_{i(t)}. \quad (\text{A1})$$

Here $s_i(t)$ is a sum of Dirac impulses; t_k is spike k 's time; δ is the Dirac delta; $g_i(t)$ is AMPA conductance; τ_{AMPA} is its decay time; w_{probe} is conductance added per spike; and t is time.

Equation A1 is the continuous-time counterpart of the decay-then-add update in Equation 3.

For stationary Poisson rate λ , the mean conductance satisfies

$$0 = -\frac{|g|_\lambda}{\tau_{\text{AMPA}}} + w_{\text{probe}}\lambda, \quad |g|_\lambda = \lambda w_{\text{probe}}\tau_{\text{AMPA}}. \quad (\text{A2})$$

Here λ is encoding rate times pixel-intensity, and $\bar{g}_\lambda$ is stationary mean conductance. The zero on the left states that the mean no longer changes with time.

Setting the mean membrane derivative in Equation 5 to zero gives

$$0 = g_{\text{L,E}}(E_L - |v|_\lambda) + |g|_\lambda(E_e - |v|_\lambda), \quad |v|_\lambda = \frac{g_{\text{L,E}}E_L + |g|_\lambda E_e}{g_{\text{L,E}} + |g|_\lambda}. \quad (\text{A3})$$

Here $\bar{v}_\lambda$ is stationary mean voltage; $g_{\text{L,E}}$ is leak conductance; and E_L and E_e are leak and AMPA reversal potentials. Other symbols follow Equation A2.

A.2 Local synapse-plus-membrane response

Write each signal as its operating point plus a small fluctuation:

$$g_i = |g|_\lambda + \delta g_i, \quad v_i = |v|_\lambda + \delta v_i, \quad s_i = \lambda + \delta s_i. \quad (\text{A4})$$

Here δg_i , δv_i , and δs_i are complete conductance, voltage, and input perturbation variables around the means defined in Equations A2 and A3.

To derive Equation A5, substitute the conductance and input decompositions from Equation A4 into the synapse equation, Equation A1:

$$\frac{d(|g|_\lambda + \delta g_i)}{dt} = -\frac{|g|_\lambda + \delta g_i}{\tau_{\text{AMPA}}} + w_{\text{probe}}(\lambda + \delta s_i).$$

The operating point is stationary, so its time derivative is zero. Expanding the right-hand side then gives

$$\frac{d\delta g_i}{dt} = \left(-\frac{|g|_\lambda}{\tau_{\text{AMPA}}} + w_{\text{probe}}\lambda \right) - \frac{\delta g_i}{\tau_{\text{AMPA}}} + w_{\text{probe}}\delta s_i.$$

The parenthesized stationary terms cancel by Equation A2. The remaining perturbation dynamics are

$$\frac{d\delta g_i}{dt} = -\frac{\delta g_i}{\tau_{\text{AMPA}}} + w_{\text{probe}}\delta s_i. \quad (\text{A5})$$

To derive Equation A6, substitute the conductance and voltage decompositions from Equation A4 into the membrane equation, Equation 5:

$$C_E \frac{d(|v|_\lambda + \delta v_i)}{dt} = g_{\text{L,E}}(E_L - |v|_\lambda - \delta v_i) + (|g|_\lambda + \delta g_i)(E_e - |v|_\lambda - \delta v_i).$$

As above, the stationary mean has zero time derivative. Expanding and grouping the right-hand side gives

$$C_E \frac{d\delta v_i}{dt} = (g_{\text{L,E}}(E_L - |v|_\lambda) + |g|_\lambda(E_e - |v|_\lambda)) - (g_{\text{L,E}} + |g|_\lambda)\delta v_i + (E_e - |v|_\lambda)\delta g_i - \delta g_i\delta v_i.$$

The first parenthesized group is zero by Equation A3. The final product is second order in the small perturbations and is omitted by the local linear approximation. This leaves

$$C_E \frac{d\delta v_i}{dt} = -(g_{L,E} + |g|_\lambda) \delta v_i + (E_e - |v|_\lambda) \delta g_i. \quad (\text{A6})$$

Here C_E is excitatory-cell capacitance. All other symbols follow Equations A1–A4.

To derive Equation A7, take the Fourier transform of Equation A5. The derivative becomes $i\omega$ times the transformed signal:

$$i\omega \delta g_{i(\omega)} = -\frac{\delta g_{i(\omega)}}{\tau_{\text{AMPA}}} + w_{\text{probe}} \delta s_{i(\omega)}.$$

Move both conductance terms to the left, then divide by their common factor:

$$\left(i\omega + \frac{1}{\tau_{\text{AMPA}}}\right) \delta g_{i(\omega)} = w_{\text{probe}} \delta s_{i(\omega)},$$

$$\delta g_{i(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \delta s_{i(\omega)}. \quad (\text{A7})$$

Equation A8 follows in the same way from Equation A6. Fourier transformation first gives

$$i\omega C_E \delta v_{i(\omega)} = -(g_{L,E} + |g|_\lambda) \delta v_{i(\omega)} + (E_e - |v|_\lambda) \delta g_{i(\omega)}.$$

Collect the voltage terms on the left:

$$(i\omega C_E + g_{L,E} + |g|_\lambda) \delta v_{i(\omega)} = (E_e - |v|_\lambda) \delta g_{i(\omega)}.$$

Dividing by the coefficient of $\delta v_i(\omega)$ gives

$$\delta v_{i(\omega)} = \frac{E_e - |v|_\lambda}{i\omega C_E + g_{L,E} + |g|_\lambda} \delta g_{i(\omega)}. \quad (\text{A8})$$

Here ω is angular frequency, i is the imaginary unit, and each argument ω denotes a Fourier-domain signal.

Equations A5–A8 form a **linear time-invariant (LTI)** system after linearization about the operating point λ . Because the system has been linearized about the operating point, it is locally linear and time-invariant, so the standard transfer-function formalism applies. The canonical frequency-domain input–output relationship is

$$\delta v_{i(\omega)} = G_{\lambda(\omega)} \delta s_{i(\omega)}.$$

Here $G_\lambda(\omega)$ is the local transfer function relating input spike-train perturbations to membrane-voltage perturbations. For non-zero input, the input–output relationship can be rearranged as

$$G_{\lambda(\omega)} = \frac{\delta v_{i(\omega)}}{\delta s_{i(\omega)}}.$$

Thus the ratio follows from the standard input–output relationship rather than serving as an independent definition. To derive $G_\lambda(\omega)$, start from Equation A7:

$$\delta g_{i(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \delta s_{i(\omega)}.$$

Substitute this expression for $\delta g_i(\omega)$ into Equation A8:

$$\delta v_{i(\omega)} = \frac{E_e - |v|_\lambda}{i\omega C_E + g_{L,E} + |g|_\lambda} \left(\frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \delta s_{i(\omega)} \right).$$

Reorder the scalar factors and factor out $\delta s_i(\omega)$:

$$\delta v_{i(\omega)} = \left(\frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - |v|_\lambda}{i\omega C_E + g_{L,E} + |g|_\lambda} \right) \delta s_{i(\omega)}.$$

Comparing this result with the canonical LTI relationship identifies the coefficient multiplying $\delta s_i(\omega)$ as the transfer function:

$$G_{\lambda(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - |v|_\lambda}{i\omega C_E + g_{L,E} + |g|_\lambda}. \quad (\text{A9})$$

Equation A9 is therefore the synaptic filter multiplied by the membrane filter, not an arbitrarily chosen ratio. Its λ -dependence captures the change in membrane gain and effective time constant with mean conductance.

A.3 Finite-window averaging

Start from the baseline-subtracted Step 1 feature in Equation 7:

$$z_i = \frac{1}{T} \int_0^T (v_{i(t)} - E_L) dt.$$

Within the stationary linearized model used in this appendix, substitute the voltage decomposition from Equation A4:

$$z_i = \frac{1}{T} \int_0^T (|v|_\lambda + \delta v_{i(t)} - E_L) dt.$$

The operating-point voltage and leak reversal potential are constant in time, so their integral is their difference multiplied by T. Dividing by T gives

$$z_i = |v|_\lambda - E_L + \frac{1}{T} \int_0^T \delta v_{i(t)} dt.$$

Define the operating-point feature and its perturbation by

$$|z|_\lambda = |v|_\lambda - E_L, \quad \delta z_i = z_i - |z|_\lambda.$$

Subtracting the operating-point feature from the expanded expression for z_i cancels the constant term and leaves

$$\delta z_i = \frac{1}{T} \int_0^T \delta v_{i(t)} dt. \quad (\text{A10})$$

Thus δz_i is the fluctuation around the operating-point feature, obtained by averaging the voltage perturbation over the 200 ms presentation duration T. To derive the frequency response, represent the averaging operation as a causal rectangular kernel $a_T(u)$, where

$$a_{T(u)} = \frac{1}{T}, \quad 0 \leq u \leq T,$$

and $a_T(u)$ is zero outside that interval. Applying this kernel as a moving-average filter gives

$$\delta z_{i(t)} = \int_{-\infty}^{\infty} a_{T(u)} \delta v_{i(t-u)} du = \frac{1}{T} \int_0^T \delta v_{i(t-u)} du.$$

At the end of the presentation, set t equal to T . To put the resulting integral into the same form as Equation A10, introduce the new variable

$$t' = T - u, \quad dt' = -du.$$

When u is zero, t' is T ; when u is T , t' is zero. The substitution therefore reverses the integration limits. Its minus sign reverses them back:

$$\delta z_{i(T)} = \frac{1}{T} \int_0^T \delta v_{i(T-u)} du = -\frac{1}{T} \int_T^0 \delta v_{i(t')} dt' = \frac{1}{T} \int_0^T \delta v_{i(t')} dt'.$$

The name of an integration variable has no effect on the value, so replacing t' with t recovers Equation A10 exactly.

For any time-domain function $f(u)$, use the Fourier-transform convention

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(u) \exp(-i\omega u) du.$$

Here $\hat{f}(\omega)$ is the Fourier transform of $f(u)$. Applying this definition with $f(u)$ equal to the rectangular kernel $a_T(u)$, and naming the result $A_T(\omega)$, gives

$$A_{T(\omega)} = \int_{-\infty}^{\infty} a_{T(u)} \exp(-i\omega u) du = \frac{1}{T} \int_0^T \exp(-i\omega u) du.$$

Evaluating the integral and rearranging gives

$$A_{T(\omega)} = \frac{\exp(-i\omega T) - 1}{-i\omega T} = \frac{1 - \exp(-i\omega T)}{i\omega T}.$$

At zero frequency, the continuous limit of $A_T(\omega)$ is one, as expected: averaging does not change a constant signal. By the convolution theorem, the averaging filter multiplies the voltage spectrum:

$$\delta z_{i(\omega)} = A_{T(\omega)} \delta v_{i(\omega)}.$$

Substitute the input-voltage relation from Equation A9:

$$\delta z_{i(\omega)} = A_{T(\omega)} G_{\lambda(\omega)} \delta s_{i(\omega)}.$$

Define the complete local input-feature response by the canonical LTI relationship

$$\delta z_{i(\omega)} = H_{\lambda(\omega)} \delta s_{i(\omega)}.$$

Comparing the coefficients multiplying $\delta s_i(\omega)$ identifies $H_{\lambda}(\omega)$. The two components of Equation A11 are therefore

$$A_{T(\omega)} = \frac{1 - \exp(-i\omega T)}{i\omega T}, \quad H_{\lambda(\omega)} = A_{T(\omega)} G_{\lambda(\omega)}. \quad (\text{A11})$$

Here $A_T(\omega)$ is the averaging-window response, $\exp$ is the exponential function, and $H_\lambda(\omega)$ is the complete response from input spike-train perturbation to averaged-voltage perturbation.

For the Bode comparison, the zero-frequency-normalized magnitude of the complete input-feature response is

$$B_{H(\omega)} = 20 \log_{10} \left(\frac{|H_{\lambda(\omega)}|}{|H_{\lambda(0)}|} \right). \quad (\text{A12})$$

Here B_H is magnitude in decibels; $|\cdot|$ is complex magnitude; and $\log_{10}$ is the base-ten logarithm. Only H_λ is plotted because it includes the averaging window and is the response used to predict the ANN feature variance.

A.4 Poisson-noise variance

A centred, unit-amplitude Poisson point process has a flat two-sided input power spectral density:

$$S_{\text{in}(\omega)} = \lambda. \quad (\text{A13})$$

Here $S_{\text{in}}(\omega)$ is input power per unit angular frequency and λ is mean spike rate. Passing this noise through Equation A11 gives

$$S_{z(\omega)} = |H_{\lambda(\omega)}|^2 S_{\text{in}(\omega)}. \quad (\text{A14})$$

Here $S_z(\omega)$ is predicted feature-noise power and $|H_\lambda(\omega)|^2$ is the transmitted fraction of input-noise power. Under the two-sided angular-frequency convention, total predicted variance is

$$\text{Var}_{\text{linear}(z)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |H_{\lambda(\omega)}|^2 S_{\text{in}(\omega)} d\omega. \quad (\text{A15})$$

Here $\text{Var}_{\text{linear}}(z)$ is predicted feature variance and π is the circle constant; the remaining symbols follow Equations A11–A14. This result is exact for the stated linear stationary model, but only approximate for the finite, discrete, conductance-dependent probe. Step 3 therefore compares it with empirical variance rather than assuming agreement.